

Diagnosis

Acute Rx

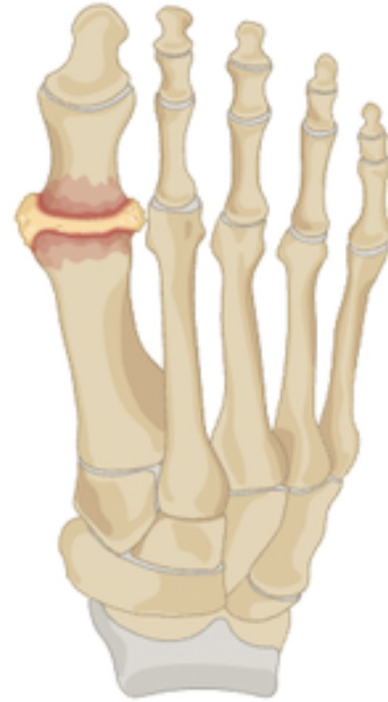
Modifiable RFs

Urate Lowering
Rx

Takeaways

Cases

Clinical Manifestations and Management of Gout



Diagnosis

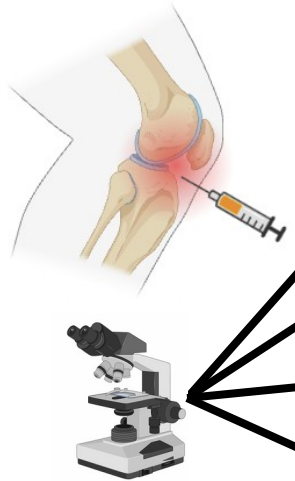
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Mimics

Gout

Septic

Pseudogout

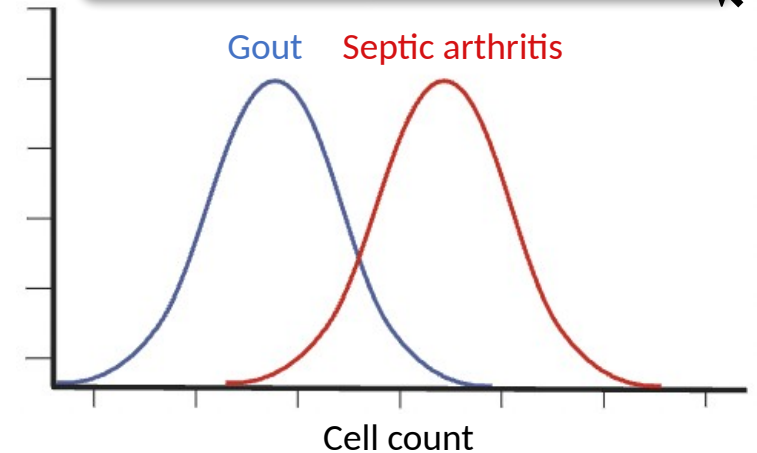
Hemarthrosis

Clinical Features

- Excruciatingly painful peripheral joint
- Abruptly red, swollen, warm
- 90% with MTP at some point
- +/- fever



Cell Count Distribution Curves



Gout:
Cell count typically 2,000- 75,000
+ monosodium urate crystals*

Septic arthritis:
Cell count typically >75,000*

Pseudogout:
- Rhomboid crystals
- More commonly larger joints
(knee involvement >50%)

Hemarthrosis:
- Overtly sanguineous fluid

- Some overlap in cell counts
- E.g., < 75,000 in septic arthritis:
 - Arthrocentesis following antibiotics
 - Gonococcal infection

Diagnosis

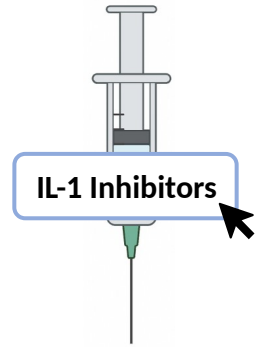
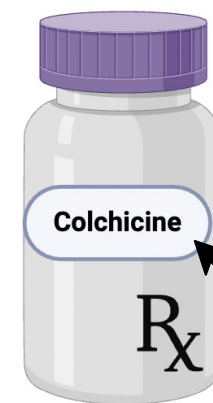
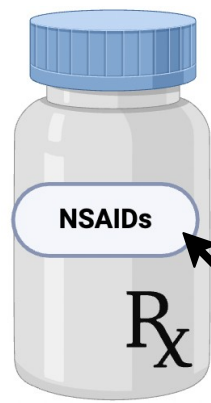
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Contra- indications	CAD Kidney disease GI bleed	* Concurrent infection * Hyperglycemia	Concurrent CYP3A4 inh Renally dosed in CKD	Concurrent infection
Dosing	Naproxen: 500mg BID Meloxicam: 15mg QD Indomethacin: 50mg TID x5-7 days	Prednisone: 40mg PO x5d Triamcinolone: Intraarticular Large joint = 40 mg (or equivalent Rx/dose)	1.2mg → 0.6mg in 1hr → 0.6mg QD or BID until flare resolves	Anakinra SQ (\$): 100 mg QD until symptoms improve, (3-5 days) Canakinumab SQ (\$\$\$): single dose 150 mg
Other Considerations	Best if <48hr from onset Equivalent efficacy of NSAIDs	Extend therapy to 7-10 days if severe or tophaceous Taper or stop	Good 'pill-in-pocket' option Avoid if onset >48 hrs GI side effects Can cause statin myotoxicity	With rheum referral Use in polyarticular gout if contraindications to alternatives

* Relative contraindications

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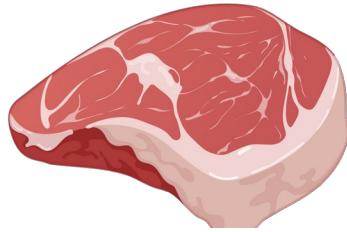
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Diet

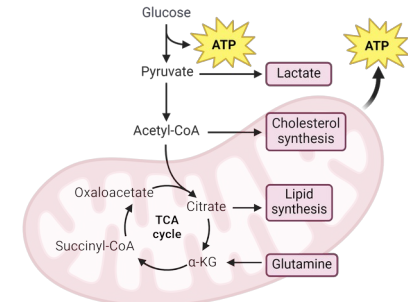
- Red meat
- Shellfish
- Alcohol: beer & spirits
- Soft drinks



Medications

- Loop & thiazides diuretics
- Beta blockers
- ACE-I & ARBs (except losartan*)
- Aspirin
- Some TB medications

*Losartan and amlodipine reduce risk



Metabolic

- Hyperlipidemia
- Hypertension

*Treating HTN/HLD reduces risk

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Indications

Rx Options

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Indications

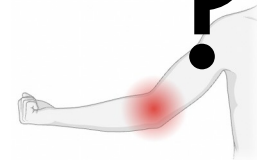
≥2 gout flares in 1 year



1 flare + erosions on X-Ray



Presence of tophi*



*Common locations: fingers,
forearms, elbows, Achilles tendons,
and the antihelices of the ear.

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Indications

Rx Options

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Xanthine Oxidase Inhibitors

Allopurinol \$

- Start 100 mg QD
- Max dose 800 mg QD
- Reduce dose for GFR<60
- **Consider HLA-B5801**

Febuxostat \$\$\$

- Start 40 mg qD
- Max dose 120 mg QD
- **Black box warning:** increased risk of CV death

Escalate dose Q2-4 wks

Goal uric acid <6 even for tophaceous gout

*SE Asian & African American
at risk of hypersensitivity

Prophylaxis

Colchicine

Low dose: 0.6 mg
QD or BID

NSAIDs

Low dose

Prednisone

Low dose: 5-10mg
QD

**MUST use prophylaxis until XOI is
at a stable dose**

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1. Starting urate lowering therapy concurrently with acute gout treatment **will not** worsen or prolong flare
2. Initiating urate-lowering therapy **without** acute gout prophylaxis can precipitate flare
3. Patients on chronic urate lowering therapy who flare up should **not** stop their urate lowering therapy
4. **Up titrate** urate lowering therapy q2-4wks to goal uric acid level **<6 mg/dL**

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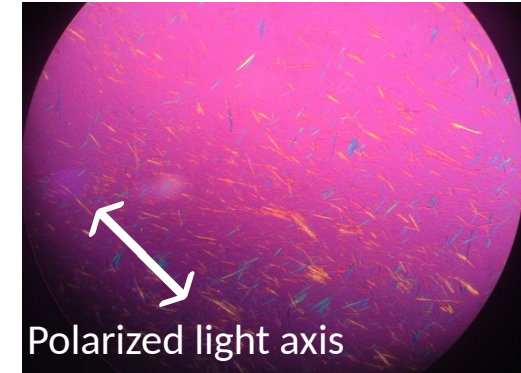
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Case 1

Mr. P is a 72 year old male with a history of insulin dependent DM2, HTN (on lisinopril) and HLD (on atorvastatin) who presents with 1 day of a hot, swollen, and painful right ankle. He denies any trauma to the area and does not endorse fever or chills.

What is the next step in diagnosis?

- XR of the swollen joint to look for fracture, erosions, etc.
- R ankle **Arthrocentesis** to rule out septic arthritis
- Gram stain/culture: negative
- Crystal analysis: negative birefringence



What is the best way to manage his acute flare?

- Short course of NSAIDs or colchicine
- History of diabetes - avoid corticosteroids if possible
- Transition ACE -> ARB

What are Mr. P's risk factors for the developing gout

Non-Modifiable

Age
Male gender

Modifiable

Metabolic disorders
(diabetes, HTN, hyperlipidemia)
ACE inhibitor use

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Case 2

Mr. K is 49 yo Han Chinese male referred for the management of recurrent flares of gouty arthritis. H/o 3 flares this year. PMHx notable for CKD and HFpEF. On exam, he has a stone-like structure on his right elbow (pictured below). His R great toe and right 3rd MCP are hot, and tender to touch.

How would you classify Mr. K's gout and presentation?

Tophaceous gout w/ acute polyarticular attack

Chronic tophus



Tophus with calcification in olecranon bursa



Courtesy of Matt Skalski,
Radiopaedia.org, rID: 29768

How would you approach management?

- Flare txt: Prednisone preferred given h/o renal disease and heart failure
- ULT: Allopurinol given # of flares, if GFR <60, dose reduce from 100 mg to 50 mg QD
- Ppx: Renally dosed colchicine vs low dose prednisone (continued for 9-12 months)

What additional lab test could you order prior to initiating allopurinol?

HLA-B5801- increased risk of allopurinol hypersensitivity syndrome due to Han Chinese background

What should Mr. K's goal uric acid level be?

<6 mg/dL

Case 3

60yo man with known tophaceous gout on allopurinol and recent treatment of cellulitis of his LLE who presents to clinic with 3 days of severe pain and swelling of his L ankle.

Vitals: Temp of 101F.

Exam: warmth, erythema, and significant pain of L ankle with exquisite TTP over joint line.

What labs do you want to order?

- CBC: WBC 20
- Blood cultures pending
- *** Uric acid may be falsely low, high, or normal during acute gout flare

What would you do next?

- Arthrocentesis: synovial fluid leukocyte count 100,000, culture pending
- Polarized microscopy- negatively birefringent monosodium urate crystals

What is your next step?

- Consider ortho consult for washout- **source control**
- ANTIBIOTICS!
- Avoid steroids in the setting of acute infection
- Consider NSAIDs or colchicine for gout, until potential wash-out



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2025 Context — Guidelines Current

ACR 2020 gout guideline remains the standard as of 2026 — no newer ACR guideline supersedes it.

Allopurinol first-line ULT for all (incl. CKD); start low, treat-to-target urate <6 mg/dL; HLA-B*5801 screening in Southeast-Asian and African-American ancestry before allopurinol.

Febuxostat: FDA boxed warning for CV/all-cause death (CARES, NEJM 2018) — second-line, reserved for allopurinol intolerance/failure. Already shown on the deck.

Newer agents: dotinurad (selective URAT1 reabsorption inhibitor) approved in Japan; as of 2026 still investigational in the US — not yet a US option. Pegloticase remains the option for refractory tophaceous gout.

Lifestyle: losartan and amlodipine have mild uricosuric benefit and are reasonable preferential BP agents in gout; do NOT stop indicated low-dose aspirin for gout (ACR 2020).

Take-Home Points

1. Tap the joint — perform arthrocentesis to rule out septic arthritis whenever there are signs of infection or an ambiguous presentation (crystals do not exclude infection).
2. Treat flares with NSAIDs, colchicine (if <48 h from onset), or steroids; match the agent to comorbidities. Anakinra (with rheumatology) if all are contraindicated.
3. Start urate-lowering therapy (allopurinol first-line) for ≥ 2 flares/yr, 1 flare + erosions, or tophi — also consider after 1 flare if CKD ≥ 3 , urate > 9 , or urolithiasis.
4. Treat-to-target: up-titrate ULT q2-4 weeks to urate < 6 mg/dL (< 5 for tophi); keep flare prophylaxis running until the target ULT dose is reached (3-6 months).
5. Screen HLA-B*5801 before allopurinol in high-risk ancestry; starting ULT during an acute flare does not worsen or prolong it.

Attribution & Changelog

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Optimized 2026-06-11 (deck: Gout_TeachIM.pptx; all 10 original interactive slides preserved).

Corrections: navigation tab label "Urate Lowing Rx" -> "Urate Lowering Rx" (all 10 slides); Case 1 (slide 8) "rule our septic arthritis" -> "rule out". Single XML runs each; all click-to-reveal animations preserved.

Currency check (June 2026): verified against ACR 2020 gout guideline (FitzGerald, Arthritis Care Res 2020) — standing teaching unchanged. CARES febuxostat boxed warning already on deck.

Companion files: none published on the lesson page. Lesson package: HTML launcher, one-pager PDF, Anki cards.